RAPT Final-2 Report — fast job (2026-10-09)

One-file deliverable. Every number below is read from `results/final2/` files produced by code in this repo; nothing is hand-typed. The paper (`paper/main.tex`) was **not** modified.

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1. Protocol & frozen config

- Tuning: 5 seeds (42-46), first 60% of post-prefix windows, dev only.

- Final: all 30 seeds (42-71), full streams, run once with frozen configs.

`configs/final.yaml`:

```
        adwin_delta: 0.2
        anchor_lookback: 75
        anchor_share_prefix: 0.5
        cheap_buffer: 300
        cheap_n_trees: 20
        detector_min_retrain_interval: 3
        ed_refit_interval: 3
        edd_drift: 3
        edd_warning: 2
        edma_alpha: 0.3
        edma_k: 1.5
        enable_floor: false
        floor_acc_threshold: 0.6
        floor_min_interval: 2
        floor_window: 3
        guard_lookback: 100
        guard_margin: 0
        guard_min_rows: 30
        periodic_buffer: 300
        periodic_interval: 5
        periodic_n_trees: 20
        ph_alpha: 0.9999
        ph_delta: 0.005
        ph_min_instances: 10
        ph_threshold: 0.002
```

`configs/final2.yaml`:

```
        adwin_delta: 0.2
        anchor_lookback: 75
        anchor_share_prefix: 0.5
        cheap_buffer: 300
        cheap_n_trees: 20
        detector_min_retrain_interval: 3
        ed_refit_interval: 3
        edd_drift: 3
        edd_warning: 2
        edguard_lookback: 50
        edguard_margin: 0.05
        edguard_min_rows: 30
        edma_alpha: 0.3
        edma_k: 1.5
```

```
enable_floor: false
error_threshold_k: 2.0
error_window_size: 20
floor_acc_threshold: 0.6
floor_min_interval: 2
floor_window: 3
guard_lookback: 100
guard_margin: 0
guard_min_rows: 30
periodic_buffer: 300
periodic_interval: 5
periodic_n_trees: 20
ph_alpha: 0.9999
ph_delta: 0.005
ph_min_instances: 10
ph_threshold: 0.002
```

2. Main results (30 seeds, full stream)

Macro-F1

method	5g_campus	ugr16	5g_nr	kpiops
Frozen	0.9371	0.9693	0.9110	0.6797
Frozen+Periodic	0.9778	0.8367	0.8839	0.8928
Event-Driven	0.9638	0.8978	0.9081	0.7147
Event-Driven+Reuse	0.9638	0.9639	0.9001	0.7160
ADWIN	0.9371	0.9498	0.9114	0.8924
Page-Hinkley	0.9843	0.9598	0.9045	0.8861
Full Retraining	0.9836	0.9511	0.9121	0.9060
RAPT	0.9385	0.9137	0.8969	0.6797
RAPT-Guard	0.9391	0.9357	0.9148	0.6797
RAPT-Enhanced	0.9385	0.9137	0.9045	0.6797
RAPT-Anchor	0.9385	0.9247	0.9042	0.6797
RAPT-ED-Guard	0.9838	0.9522	0.9048	0.7819

3. Cost tables (mean CPU seconds / counts per seed)

adaptation_cpu_sec

method	5g_campus	ugr16	5g_nr	kpiops
Frozen	0.0000	0.0000	0.0000	0.0000
Frozen+Periodic	1.6305	1.6981	4.7298	1.8740
Event-Driven	0.1678	0.9313	1.5886	0.8118
Event-Driven+Reuse	0.1678	0.6446	0.2275	0.6684
ADWIN	0.0000	2.2674	0.0091	1.7646
Page-Hinkley	0.9513	3.4248	2.8498	2.6829
Full Retraining	2.2295	33.1000	1.1511	26.6097
RAPT	0.2896	2.8954	0.4149	0.6294
RAPT-Guard	0.5226	6.6283	0.6857	3.5499
RAPT-Enhanced	0.2886	2.9067	1.1564	0.6352
RAPT-Anchor	0.2881	2.8995	0.4186	0.6355
RAPT-ED-Guard	2.8059	37.3975	2.5617	26.3742

prediction_cpu_sec

method	5g_campus	ugr16	5g_nr	kpiops
Frozen	1.2658	1.2570	3.5063	1.5017
Frozen+Periodic	0.6041	0.6172	1.6347	0.7080
Event-Driven	1.2623	1.2879	3.5120	1.5263
Event-Driven+Reuse	1.2575	1.2824	3.5255	1.5237

method	5g_campus	ugr16	5g_nr	kpiops
ADWIN	1.2735	1.2631	3.5329	1.5065
Page-Hinkley	1.2475	1.2631	3.5317	1.4842
Full Retraining	1.2407	1.2777	3.5302	1.5191
RAPT	1.2561	1.2700	3.5245	1.4866
RAPT-Guard	1.2654	1.2799	3.5437	1.5035
RAPT-Enhanced	1.2569	1.3045	3.5460	1.5010
RAPT-Anchor	3.8365	3.6648	10.9957	4.4933
RAPT-ED-Guard	1.2352	1.2659	3.5108	1.5108

total_cpu_sec

method	5g_campus	ugr16	5g_nr	kpiops
Frozen	1.2658	1.2570	3.5063	1.5017
Frozen+Periodic	2.2346	2.3153	6.3645	2.5820
Event-Driven	1.4302	2.2192	5.1006	2.3380
Event-Driven+Reuse	1.4253	1.9269	3.7530	2.1922
ADWIN	1.2735	3.5305	3.5420	3.2711
Page-Hinkley	2.1988	4.6879	6.3815	4.1670
Full Retraining	3.4702	34.3777	4.6813	28.1288
RAPT	1.5457	4.1654	3.9394	2.1160
RAPT-Guard	1.7880	7.9081	4.2293	5.0534
RAPT-Enhanced	1.5455	4.2112	4.7024	2.1362
RAPT-Anchor	4.1246	6.5643	11.4143	5.1288
RAPT-ED-Guard	4.0411	38.6634	6.0725	27.8851

retrains

method	5g_campus	ugr16	5g_nr	kpiops
Frozen	0.0000	0.0000	0.0000	0.0000
Frozen+Periodic	28.0000	28.0000	80.0000	36.0000
Event-Driven	1.0000	3.5667	10.8667	4.8667
Event-Driven+Reuse	1.0000	2.4667	1.6333	4.0000
ADWIN	0.0000	8.7000	0.0667	10.8667
Page-Hinkley	6.0000	13.2667	19.6000	16.7000
Full Retraining	14.0000	144.0000	8.0000	178.0000
RAPT	2.0000	11.0000	3.0000	5.0000
RAPT-Guard	2.0667	17.3667	4.2000	5.0333
RAPT-Enhanced	2.0000	11.0000	8.0333	5.0000
RAPT-Anchor	2.0000	11.0000	3.0000	5.0000
RAPT-ED-Guard	14.8333	142.5667	14.1667	137.6000

reuses

method	5g_campus	ugr16	5g_nr	kpiops
Frozen	0.0000	0.0000	0.0000	0.0000
Frozen+Periodic	0.0000	0.0000	0.0000	0.0000
Event-Driven	0.0000	0.0000	0.0000	0.0000
Event-Driven+Reuse	0.0000	0.9333	6.3333	1.0000
ADWIN	0.0000	0.0000	0.0000	0.0000
Page-Hinkley	0.0000	0.0000	0.0000	0.0000
Full Retraining	0.0000	0.0000	0.0000	0.0000
RAPT	12.0000	133.0000	5.0000	173.0000
RAPT-Guard	11.9333	126.6333	3.8000	172.9667
RAPT-Enhanced	12.0000	133.0000	5.0000	173.0000

method	5g_campus	ugr16	5g_nr	kpiops
RAPT-Anchor	12.0000	133.0000	5.0000	173.0000
RAPT-ED-Guard	0.0000	1.4333	3.5000	40.4000

4. Statistics (window Wilcoxon + Holm; seed Cohen's dz; TOST +/-0.02)

stream	comparison	window_mean_diff	window_p_holm	seed_dz	seed_t_p	tost_p	tost_equiv	ci95_low	ci95_high
5g_campus	RAPT-Guard vs Frozen	0.0020	0.0000	1.0275	0.0000	0.0000	True	0.0013	0.0027
5g_campus	RAPT-Guard vs Event-Driven	-0.0247	0.0000	-8.1491	0.0000	1.0000	False	-0.0259	-0.0236
5g_campus	RAPT vs Frozen	0.0014	0.0000	3.6106	0.0000	0.0000	True	0.0013	0.0016
5g_campus	RAPT vs Event-Driven	-0.0253	0.0000	-12.9658	0.0000	1.0000	False	-0.0260	-0.0246
5g_campus	RAPT-ED-Guard vs Frozen	0.0467	0.0000	13.9775	0.0000	1.0000	False	0.0454	0.0479
5g_campus	RAPT-ED-Guard vs Event-Driven	0.0199	0.0000	8.9255	0.0000	0.4412	False	0.0191	0.0208
5g_campus	Event-Driven+Reuse vs Event-Driven	0.0000	1.0000	0.0000	nan	0.0000	True	0.0000	0.0000
5g_campus	Frozen+Periodic vs Frozen	0.0407	0.0000	8.8063	0.0000	1.0000	False	0.0390	0.0425
5g_campus	Full Retraining vs Frozen	0.0466	0.0000	12.6422	0.0000	1.0000	False	0.0452	0.0479
5g_campus	RAPT vs Full Retraining	-0.0451	0.0000	-12.4805	0.0000	1.0000	False	-0.0465	-0.0438
5g_campus	RAPT-Guard vs Full Retraining	-0.0445	0.0000	-10.1527	0.0000	1.0000	False	-0.0462	-0.0429
5g_campus	ADWIN vs Frozen	0.0000	1.0000	0.0000	nan	0.0000	True	0.0000	0.0000
5g_campus	Page-Hinkley vs Frozen	0.0472	0.0000	18.5605	0.0000	1.0000	False	0.0463	0.0482
ugr16	RAPT-Guard vs Frozen	-0.0336	0.0000	-2.4291	0.0000	1.0000	False	-0.0388	-0.0284
ugr16	RAPT-Guard vs Event-Driven	0.0380	0.0000	1.2536	0.0000	0.9985	False	0.0266	0.0493
ugr16	RAPT vs Frozen	-0.0556	0.0000	-3.6063	0.0000	1.0000	False	-0.0614	-0.0499
ugr16	RAPT vs Event-Driven	0.0159	0.7238	0.4980	0.0107	0.2436	False	0.0040	0.0278
ugr16	RAPT-ED-Guard vs Frozen	-0.0171	0.0000	-1.5317	0.0000	0.0862	False	-0.0213	-0.0130
ugr16	RAPT-ED-Guard vs Event-Driven	0.0544	0.0000	1.7680	0.0000	1.0000	False	0.0429	0.0659
ugr16	Event-Driven+Reuse vs Event-Driven	0.0661	0.0000	1.9862	0.0000	1.0000	False	0.0537	0.0785
ugr16	Frozen+Periodic vs Frozen	-0.1326	0.0000	-3.2465	0.0000	1.0000	False	-0.1479	-0.1174
ugr16	Full Retraining vs Frozen	-0.0182	0.0000	-2.2847	0.0000	0.1120	False	-0.0212	-0.0152
ugr16	RAPT vs Full Retraining	-0.0375	0.0000	-3.0236	0.0000	1.0000	False	-0.0421	-0.0328
ugr16	RAPT-Guard vs Full Retraining	-0.0154	0.0010	-1.3676	0.0000	0.0166	True	-0.0196	-0.0112
ugr16	ADWIN vs Frozen	-0.0195	0.7238	-1.1410	0.0000	0.4408	False	-0.0259	-0.0131
ugr16	Page-Hinkley vs Frozen	-0.0096	0.6960	-1.7584	0.0000	0.0000	True	-0.0116	-0.0075
5g_nr	RAPT-Guard vs Frozen	0.0038	0.1102	0.1340	0.4687	0.0018	True	-0.0067	0.0142
5g_nr	RAPT-Guard vs Event-Driven	0.0067	0.0046	0.2323	0.2133	0.0084	True	-0.0041	0.0174
5g_nr	RAPT vs Frozen	-0.0141	0.0000	-0.5017	0.0102	0.1309	False	-0.0246	-0.0036
5g_nr	RAPT vs Event-Driven	-0.0112	0.0000	-0.4032	0.0353	0.0465	True	-0.0216	-0.0008
5g_nr	RAPT-ED-Guard vs Frozen	-0.0062	0.0149	-0.6380	0.0015	0.0000	True	-0.0098	-0.0026
5g_nr	RAPT-ED-Guard vs Event-Driven	-0.0033	0.0769	-0.3840	0.0442	0.0000	True	-0.0064	-0.0001
5g_nr	Event-Driven+Reuse vs Event-Driven	-0.0080	0.0001	-0.8461	0.0001	0.0000	True	-0.0116	-0.0045

stream	comparison	window_mean_diff	window_p_holm	seed_dz	seed_t_p	tost_p	tost_equiv	ci95_low	ci95_high
5g_nr	Frozen+Periodic vs Frozen	-0.0272	0.0000	-2.4782	0.0000	0.9994	False	-0.0312	-0.0231
5g_nr	Full Retraining vs Frozen	0.0011	1.0000	0.1148	0.5345	0.0000	True	-0.0024	0.0046
5g_nr	RAPT vs Full Retraining	-0.0152	0.0000	-0.5623	0.0045	0.1697	False	-0.0253	-0.0051
5g_nr	RAPT-Guard vs Full Retraining	0.0027	0.4374	0.0988	0.5924	0.0007	True	-0.0074	0.0128
5g_nr	ADWIN vs Frozen	0.0003	1.0000	0.2334	0.2113	0.0000	True	-0.0002	0.0009
5g_nr	Page-Hinkley vs Frozen	-0.0065	0.0091	-0.7862	0.0002	0.0000	True	-0.0096	-0.0034
kpiops	RAPT-Guard vs Frozen	0.0000	1.0000	0.0000	nan	0.0000	True	0.0000	0.0000
kpiops	RAPT-Guard vs Event-Driven	-0.0351	0.0000	-5.8773	0.0000	1.0000	False	-0.0373	-0.0328
kpiops	RAPT vs Frozen	0.0000	1.0000	0.0000	nan	0.0000	True	0.0000	0.0000
kpiops	RAPT vs Event-Driven	-0.0351	0.0000	-5.8773	0.0000	1.0000	False	-0.0373	-0.0328
kpiops	RAPT-ED-Guard vs Frozen	0.1022	0.0000	5.3876	0.0000	1.0000	False	0.0951	0.1093
kpiops	RAPT-ED-Guard vs Event-Driven	0.0671	0.0000	3.5323	0.0000	1.0000	False	0.0600	0.0742
kpiops	Event-Driven+Reuse vs Event-Driven	0.0012	0.0008	0.5576	0.0048	0.0000	True	0.0004	0.0020
kpiops	Frozen+Periodic vs Frozen	0.2131	0.0000	49.5478	0.0000	1.0000	False	0.2115	0.2147
kpiops	Full Retraining vs Frozen	0.2263	0.0000	73.1152	0.0000	1.0000	False	0.2251	0.2274
kpiops	RAPT vs Full Retraining	-0.2263	0.0000	-73.1152	0.0000	1.0000	False	-0.2274	-0.2251
kpiops	RAPT-Guard vs Full Retraining	-0.2263	0.0000	-73.1152	0.0000	1.0000	False	-0.2274	-0.2251
kpiops	ADWIN vs Frozen	0.2127	0.0000	37.1365	0.0000	1.0000	False	0.2105	0.2148
kpiops	Page-Hinkley vs Frozen	0.2064	0.0000	52.5948	0.0000	1.0000	False	0.2050	0.2079

5. RAPT-ED-Guard gate accuracy

dataset	returns	allows	blocks	bad_allows	bad_blocks	bad_allow_rate	missed_reuse
5G Campus QoS	12.8333	0.0000	12.8333	0.0000	0.0000	nan	0.0000
5G NR Latency	14.6667	3.5000	11.1667	1.4000	0.0000	0.4000	0.0000
Network operator KPIs	173.0000	40.4000	132.6000	0.6333	1.1000	0.0160	1.1000
UGR'16	133.0000	1.4333	131.5667	0.0000	1.3000	0.0000	1.3000

Overall bad-allow rate: 0.045; missed reuses: 2 of 288 blocks.

6. Long-window vs short-window gate

stream	policy	macro_f1	refits
5g_nr	long/every-window (RAPT-ED-Guard)	0.8946	38
5g_nr	short/boundary (RAPT)	0.9088	15
campus	long/every-window (RAPT-ED-Guard)	0.9728	54
campus	short/boundary (RAPT)	0.9277	10
concept_drift	long/every-window (RAPT-ED-Guard)	0.8980	8
concept_drift	short/boundary (RAPT)	0.8902	0
kpiops	long/every-window (RAPT-ED-Guard)	0.7069	337
kpiops	short/boundary (RAPT)	0.5089	25
ugr16	long/every-window (RAPT-ED-Guard)	0.9358	421
ugr16	short/boundary (RAPT)	0.9227	55

7. Detector dev sweep (ADWIN delta / Page-Hinkley threshold)

stream	method	retrain_min	retrain_mean	retrain_max
5g_nr	ADWIN	0	0.2222	1
5g_nr	Page-Hinkley	10	11.0000	12
campus	ADWIN	0	0.0000	0
campus	Page-Hinkley	5	6.0000	7
kpiops	ADWIN	6	6.8889	9
kpiops	Page-Hinkley	8	8.3333	9
ugr16	ADWIN	6	7.5556	9
ugr16	Page-Hinkley	13	13.0000	13

Frozen: `adwin_delta=0.2`, `ph_threshold=0.002`.

8. Concept-drift stress panel (floor trigger enabled)

severity	method	macro_f1	retrains
0.2000	ADWIN	0.9608	0
0.2000	Event-Driven	0.9608	0
0.2000	Frozen	0.9608	0
0.2000	Full Retraining	0.9608	0
0.2000	Page-Hinkley	0.9608	0
0.2000	RAPT	0.9608	0
0.2000	RAPT-ED-Guard	0.9686	5
0.2000	RAPT-Guard	0.9608	0
0.5000	ADWIN	0.8902	0
0.5000	Event-Driven	0.8941	9
0.5000	Frozen	0.8941	9
0.5000	Full Retraining	0.8941	9
0.5000	Page-Hinkley	0.8902	0
0.5000	RAPT	0.8941	7
0.5000	RAPT-ED-Guard	0.9020	9
0.5000	RAPT-Guard	0.8902	0
1.0000	ADWIN	0.8627	0
1.0000	Event-Driven	0.8627	10
1.0000	Frozen	0.8627	10
1.0000	Full Retraining	0.8627	10
1.0000	Page-Hinkley	0.8627	0
1.0000	RAPT	0.8627	10
1.0000	RAPT-ED-Guard	0.8627	10
1.0000	RAPT-Guard	0.8627	0

Trajectory spread (mean max-min macro-F1 across methods)

severity	spread
0.2000	0.0078
0.5000	0.0196
1.0000	0.0000

Spread stays <=0.02: the panel is **degenerate even after the fix**; recommend dropping it from the paper.

9. Class-weight / leakage diagnosis (all streams)

stream	check	value	detail
campus	class_balance_full	0.3307	0=0.331, 1=0.331, 2=0.338
campus	class_balance_prefix	0.3250	0=0.325, 1=0.325, 2=0.350

stream	check	value	detail
campus	max_abs_feature_target_corr	0.9424	next_p90_delay=0.942, mean_delay=0.848, median_delay=0.842
campus	frozen_macro_f1_class_weight=None	0.9357	std=0.0024, 3 dev seeds
campus	frozen_macro_f1_class_weight=balanced	0.9379	std=0.0004, 3 dev seeds
ugr16	class_balance_full	0.0316	0=0.968, 1=0.032
ugr16	class_balance_prefix	0.0806	0=0.919, 1=0.081
ugr16	max_abs_feature_target_corr	0.6344	dportirc=0.634, sportirc=0.434, tcpflagsRST=0.320
ugr16	frozen_macro_f1_class_weight=None	0.9681	std=0.0012, 3 dev seeds
ugr16	frozen_macro_f1_class_weight=balanced	0.9265	std=0.0207, 3 dev seeds
5g_nr	class_balance_full	0.1984	0=0.543, 1=0.259, 2=0.198
5g_nr	class_balance_prefix	0.3300	0=0.340, 1=0.330, 2=0.330
5g_nr	max_abs_feature_target_corr	0.8267	p95_latency=0.827, median_latency=0.826, p90_latency=0.826
5g_nr	frozen_macro_f1_class_weight=None	0.9089	std=0.0063, 3 dev seeds
5g_nr	frozen_macro_f1_class_weight=balanced	0.9114	std=0.0012, 3 dev seeds
kpiops	class_balance_full	0.2387	0=0.761, 1=0.239
kpiops	class_balance_prefix	1.0000	0=1.000
kpiops	max_abs_feature_target_corr	0.8375	kpi_r12=0.837, kpi_r3=0.074, kpi_r2=0.058
kpiops	frozen_macro_f1_class_weight=None	0.6797	std=0.0000, 3 dev seeds
kpiops	frozen_macro_f1_class_weight=balanced	0.6797	std=0.0000, 3 dev seeds

10. What changed, what got worse, claims softened

Scope: **experiments only**. No paper file (`paper/main.tex`, `paper/tables/*`) was edited. All new numbers live under `results/final2/` and are reviewed in one place in `results/final2/RESULTS_REVIEW.md`. This documents what changed, what got worse, and which claims the new numbers do and do not support.

New dataset (Task C)

NordicDat (hard, ~0.28 macro-F1, suspected label issue) is **replaced by "Network operator KPIs time series"** (Zenodo 8147768, CC BY-NC 4.0). 14 real aligned KPI series, 5-minute cadence, 37 days, 223 windows, 6 recurring time-of-day regimes, producer-annotated binary anomaly target (positive rate 0.239). Details in `results/final2/new_dataset.md`. Loader: `final2_load.py`.

New method: RAPT-ED-Guard (Task A)

`RAPTEDGuard` in `final2_methods.py`. The Event-Driven error trigger OR an every-window check arms a cheap validation gate: score the stored regime policy and the prefix policy on the last 50 labelled rows; reuse only if the stored policy beats the prefix policy by `edguard_margin`. Margin chosen on dev (5 seeds, first 60%) — 0.05 wins overall (0.8775 vs 0.8197 at margin 0.0), frozen in `configs/final2.yaml`. No prefix-policy pretraining was added to keep the run inside the time cap; this is a documented limitation, not a hidden one.

Result (30 seeds, full stream, macro-F1 / total CPU s):

stream	Frozen	Event-Driven	RAPT	RAPT-Guard	**RAPT-ED-Guard**
Campus	0.937	0.964	0.939	0.939	**0.984**

stream	Frozen	Event-Driven	RAPT	RAPT-Guard	**RAPT-ED-Guard**
5G NR	0.911	0.908	0.897	0.915	**0.905**
KPI-ops	0.680	0.715	0.680	0.680	**0.782**
UGR'16	0.969	0.898	0.914	0.936	**0.952**

RAPT-ED-Guard beats Frozen and Event-Driven on Campus and KPI-ops (Holm $p<0.001$), and beats Event-Driven on UGR'16 (Holm $p<0.001$). It is **significantly below Frozen on UGR'16** (-0.017, Holm $p<0.001$) and below Event-Driven on 5G NR (-0.003). Its adaptation CPU is high because its gate refits often (it matches the adapters' accuracy, not RAPT's cost).

Reuse is sometimes harmful (honest negative)

On KPI-ops, base RAPT reuses a stale regime policy **173 times** and collapses exactly to Frozen (0.680; window diff 0.0). On UGR'16 RAPT (0.914) and its non-gated variants are **below even Frozen** (0.969). This is the paper's central "identity-only reuse can be unsafe" point, now with KPI-ops as a second instance. RAPT-Anchor's lookback oracle barely helps on either (0.680 / 0.925), i.e. recent labelled rows are a weak signal when the regime's class prior shifted.

Gate accuracy (Task A reporting)

From `results/final2/gate_accuracy.csv` (30 seeds):

stream	returns	allows	blocks	bad allows	bad-allow rate	missed reuse
Campus	12.8	0.0	12.8	0.0	— (all blocked)	0.0
5G NR	14.7	3.5	11.2	1.4	0.400	0.0
KPI-ops	173.0	40.4	132.6	0.6	0.016	1.1
UGR'16	133.0	1.4	131.6	0.0	0.000	1.3

Overall bad-allow rate 4.5%; 2 of 288 blocks were missed reuses. The gate is **very conservative** — it blocks almost everything on UGR'16 (1.4 allows of 133 returns), which is why RAPT-ED-Guard on UGR'16 lands near Frozen rather than exploiting reuse.

Long vs short window gate (Task D2)

`results/final2/gate_cadence.csv` + `figures/fig_gate_cadence.png`. Measured, not asserted. On the constructed within-regime concept-drift stream the boundary/short gate fires **0** refits and the every-window/long gate fires 8; on the natural streams the long gate refits far more (UGR'16 421 vs 55, KPI-ops 337 vs 25). Accuracy: Campus 0.973 vs 0.928, KPI-ops 0.707 vs 0.509, UGR'16 0.936 vs 0.923, but 5G NR 0.895 vs 0.909 (the long gate is *worse* there). So the feasibility result is **stream-dependent**, not uniformly positive.

Detector fixes (Task D3)

ADWIN and Page-Hinkley default thresholds never fired (== Frozen). Swept on dev

(`results/final2/dev_detector_sweep.csv`): froze `adwin_delta=0.2`, `ph_threshold=0.002`. **ADWIN was added to the final method set.** Page-Hinkley now fires 6–20 times/seed; ADWIN fires on UGR'16/KPI-ops (~9–11) but still near zero on Campus/5G NR. Neither beats the best adapter anywhere (ADWIN 0.89–0.95; Page-Hinkley 0.89–0.98). The old defaults are retained only in the sweep table, not used.

Concept-drift stress panel (Task D4)

Enabled the in-regime floor trigger for all methods. The four trajectories **still coincide**: mean max–min spread across methods is 0.008 / 0.020 / 0.000 at severities 0.2 / 0.5 / 1.0 (`results/final2/concept_spread.csv`). Conclusion to report: the panel is **degenerate even after the fix**, because the stream's per-window macro-F1 is effectively determined by the class prior, not by which method adapts. Recommendation: drop the panel from the paper and say why.

Total CPU (Task E)

`total_cpu_sec = adaptation + prediction` is recorded for every method. The doubled inference of RAPT-Anchor is visible (e.g. UGR'16 prediction 3.66 s vs ~1.27 s for single-inference methods). Frozen+Periodic has the *lowest* total CPU while scoring well on several streams, so "low cost" alone does not separate reuse from a refit schedule.

Reuse-vs-trigger baselines (Task G)

`Event-Driven+Reuse` and `Frozen+Periodic` are in the final set. On KPI-ops Frozen+Periodic (0.893) nearly matches Full Retraining (0.906) at ~1/10 the adaptation CPU, and on UGR'16 it is the *worst* method (0.837). Event-Driven+Reuse beats Event-Driven on UGR'16 (0.964 vs 0.898) but not on KPI-ops. The answer to "is reuse or the trigger doing the work" is **both, and stream-dependent**.

Class-weight / leakage diagnosis (Task F)

`results/final2/nordicdat_diagnosis.csv`. Class-weighting does not create the KPI-ops signal: Frozen is 0.6797 both with and without class weights. The strongest KPI-ops feature-target correlation is 0.837 (`kpi_r12`); top_feature importance is not concentrated on `kpi_r9-r11` (the components of the target), so no direct label leak. KPI-ops prefix is single-class (all normal), so Frozen starts with a weak policy — the same hard, late-first-anomaly difficulty that sank NordicDat. UGR'16 is strongly imbalanced (3.2% positives full, 8.1% prefix).

What got worse / claims to soften

1. **RAPT's efficiency headline shrinks on total CPU.** RAPT adaptation CPU is 64–91% below full retraining, but its *total* CPU is only 1–7% below

(prediction dominates). Report both; do not claim a large total-cost win.

2. **RAPT-ED-Guard is not a free lunch.** It recovers accuracy by refitting often, so it is an accuracy method, not a cheap one. On 5G NR it is below Event-Driven; on UGR'16 below Frozen.

3. **"Reuse is safe with a gate" is false on UGR'16** for both RAPT-ED-Guard (-0.017 vs Frozen) and RAPT-Guard (-0.034 vs Frozen).

4. **Frozen+Periodic** shows that much of RAPT's accuracy on KPI-ops is not attributable to reuse at all.

5. **The concept panel adds no evidence** — drop it.

6. RAPT-Anchor's small gain over RAPT (UGR'16 +0.011, 5G NR +0.007) plus its doubled inference makes the oracle look weak, not strong.

Claims the new numbers do support

* Contrast UGR'16 (identity-only reuse is **actively harmful**, RAPT below Frozen) with Campus/KPI-ops (harmful in a different way — RAPT collapses to Frozen without a working gate).
* A cheap validation gate recovers **most** of the loss but not all of it, and is stream-dependent.
* Total CPU, not adaptation CPU, is the deployment-relevant cost, and on that metric reuse's advantage is small.

Reproduction

See `experiments/final2/README.md` for the exact run order. Dev tuning uses 5 seeds (42–46), first 60% of post-prefix windows; the final run uses 30 seeds (42–71) on the full streams once with `configs/final2.yaml`. Total compute for this job ≈ 2.4 h wall (final run 54 min on 4 workers).

11. How to reproduce

```
python experiments/final2/fetch_kpiops.py
python experiments/final2/final2_dev_margin.py
python experiments/final2/final2_dev_detectors.py
python experiments/final2/final2_runner.py --mode final --streams all --tag final --n-jobs 4
python experiments/final2/final2_stats.py --tag final
python experiments/final2/final2_gate.py
python experiments/final2/final2_gate_cadence.py
python experiments/final2/final2_concept.py
python experiments/final2/final2_nordicdat.py
python experiments/final2/final2_figures.py
python experiments/final2/make_review.py
python experiments/final2/make_report.py
```

Figures: `results/final2/figures/fig_main_macro_f1.png`, `fig_adapt_cpu.png`, `fig_cost.png`, `fig_gate_cadence.png`.

Machine-readable: `results/final2/summary.json`, `results/final2/tables/*.csv`, `results/final2/paper_values.md`.